

MD RAYHANUL ISLAM

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SUMMARY

PhD student specializing in Time-Sensitive Networking (TSN) and deterministic communication, with hands-on experience designing, implementing, and evaluating TSN scheduling systems on real hardware testbeds (TSN switches and edge compute nodes). Background spans network scheduling, IoT/edge systems, formal verification, and deep reinforcement learning. Seeking internships and full-time roles in deterministic / time-sensitive networking, network scheduling, and edge systems.

RESEARCH INTEREST

Time-Sensitive Networking, Network Scheduling, IoT, Deep Learning, and Formal Verification.

EDUCATION

PhD Student, University of Connecticut Fall 2024 - Current
CGPA: 4.0 (till now), and Relevant Coursework: Computer Architecture, Program Analysis for Software Security, Reinforcement Learning, Architecture of IoT, and Time-Sensitive Networking.

Research Area: Time-Sensitive Networking, Network Scheduling, Formal Verification, Deep Reinforcement Learning.

Advisor: Dr. Song Han

Master of Science in Computer Science, Iowa State University Spring 2024
CGPA: 3.82 and Relevant Coursework: Machine Learning, Introduction to Machine Learning, Principles of Artificial Intelligence, Natural Language Processing, Design and Analysis of Algorithms, Theory of Computation, Formal Methods in Software Engineering, Program Analysis, Motion Planning for Robotics and Autonomous Systems, and Advanced Topics in Autonomous Systems and Robotics.

Thesis: Quantifying Safety in the Partially Observable Environment: Application to Autonomous Vehicles. *Advisor: Dr. Tichakorn Wongpiromsarn*

Master of Science in Software Engineering, University of Dhaka
CGPA: 3.96.

Bachelor of Science in Software Engineering, University of Dhaka
CGPA: 3.86.

EXPERIENCE

Graduate Research Assistant Fall 2024 – Present
School of Computing, University of Connecticut Storrs, CT

- Contribute to the NSF-funded project *Collaborative Research: CSR: Small: Accurate and Private Multi-Camera Surveillance System on Time-Sensitive Networks*, focusing on deterministic communication, TSN scheduling, run-time network reconfiguration, and privacy-aware multi-camera edge intelligence.
- Design and implement TSN-based edge-AI testbed components integrating object detection, privacy-aware transmission, and deterministic networking.
- Teach and assist in **Artificial Intelligence (Spring 2026)**.

Research Intern June 16, 2025 - August 06, 2025
College of Engineering and Computer Science, University of Central Florida Orlando, FL

- Contributed to implementing a hardware fuzzer to expose micro-architectural side-channel attacks.

Graduate Research & Teaching Assistant Fall 2021 – Spring 2024
Autonomous System Lab, Department of Computer Science, Iowa State University Ames, IA

- Contributed to the NSF-funded project *Sharing the World with Autonomous Systems: What Goes Wrong and How to Fix It*, focusing on safety analysis of interacting autonomous systems in partially observable environments.

- Developed a framework to analyze perception and decision-making inconsistencies using formal verification techniques.
- Taught and assisted in Object-Oriented Programming, Principles of Programming Languages, Principles of Artificial Intelligence, and Computer Science Senior Design Project.

Lecturer

Institute of Leather Engineering & Technology, University of Dhaka

Sep 2017 - Aug 2021
Dhaka, Bangladesh

Software Engineer

Kaz Software Ltd.

May 2015 - Sep 2017
Dhaka, Bangladesh

- Worked on the back-end and front-end of multiple projects.
- Used languages and frameworks: C#, Java, ASP.NET, JavaScript, Angular Js, Knockout Js, Ext Js, etc.

Internship

Kaz Software Ltd.

Jan 2013 - Jul 2013
Dhaka, Bangladesh

PUBLICATIONS

- Md Rayhanul Islam, Tichakorn Wongpiromsarn, and Samik Basu. "Analysis of the Effects of Perception Inconsistency Among Interacting Vehicles in Partially Observable Environments" In 2024 27th IEEE International Conference on Intelligent Transportation Systems (ITSC).
- Atish Kumar Dipongkor, Rayhanul Islam, Md Shafiuzzaman, Md Asif Nashiry, Syed Md Galib, and Khaza Moinuddin Mazumder. "AcPgChecker: Detection of Plagiarism among Academic and Scientific Writings." In 2021 Joint 10th International Conference on Informatics, Electronics & Vision (ICIEV) and 2021 5th International Conference on Imaging, Vision & Pattern Recognition (icIVPR), pp. 1-6. IEEE, 2021.
- Atish Kumar Dipongkor, Iftekhar Ahmed, Rayhanul Islam, Nadia Nahar, Abdus Satter, and Md Saeed Siddik. "ABMMRS Eradicator: Improving Accuracy in Recommending Move Methods for Web-based MVC Projects and Libraries Using Method's External Dependencies." International Journal of Software Engineering and Knowledge Engineering 30, no. 09 (2020): 1289-1307.
- Rayhanul Islam, Abdus Satter, Atish Kumar Dipongkor, Md Saeed Siddik, and Kazi Sakib. "A Novel Approach for Converting N-Dimensional Dataset into Two Dimensions to Improve Accuracy in Software Defect Prediction." J. Softw. 15, no. 6 (2020): 147-162.
- Kishan Kumar Ganguly, Md Saeed Siddik, Rayhanul Islam, and Kazi Sakib. "An environment aware learning-based self-adaptation technique with reusable components." International Journal of Modern Education and Computer Science 11, no. 6 (2019): 53.
- Rayhanul Islam, and Kazi Sakib. "A package-based clustering approach to enhance the accuracy and performance of software defect prediction." International Journal of Software Engineering, Technology and Applications 2, no. 1 (2017): 1-21.

PRESENTATIONS AND TALKS

Analysis of the Effects of Perception Inconsistency Among Interacting Vehicles in Partially Observable Environments (Online) **09/2024**

- Presented at the 27th IEEE International Conference on Intelligent Transportation Systems (ITSC 2024), September 24-27, 2024, in Edmonton, Canada.

FELLOWSHIPS AND AWARDS

Pratt & Whitney Institute for Advanced Systems Engineering (PW-IASE) Graduate Fellowship, UConn 2026–2027

- Provides fellowship support equivalent to 50% of a Graduate Assistantship stipend for Fall 2026 to support doctoral research.

UConn Computer Science Predoctoral Fellowship, School of Computing, UConn

May 2026

- Awarded in recognition of academic performance, research progress, and promise as a Ph.D. researcher.

PROJECTS

Scheduling of Computing and networking research

Fall 2025

Project for CSE 5312: Architecture of IoT

- This project builds an SMT model to schedule the synchronization of the computing and networking resources. This has been tested on a testbed with a TSN switch and Raspberry Pi 4 computing nodes.

AWS Deepracer

Spring 2025

Project for CSE 5802: Machine Learning (Reinforcement Learning)

- This project develops a deep reinforcement learning model trained on the AWS Deepracer platform for the AWS Deepracer competition.

RISC-V Simulator: Dynamic Branch Prediction and Data Cache

Fall 2024

Project for CSE 5302: Computer Architecture

- This project implements three dynamic branch predictors (static, 1-level, and 2-level) and three data caching approaches (Direct Mapped cache, 2-way set associative, and 4-way set associative) on the RISC-V simulator.

Custom Query Answering with RAG and Pre-Trained Language Models

Spring 2024

Project for COM S 579: Natural Language Processing

- This project develops a Retrieval augmented generation (RAG) system using a pre-trained large language model to provide any customized answer to a query based on a given document.

Policy Evaluation on Uncertain Environment

Fall 2023

Project for COM S 676X: Advanced Topics in Autonomous Systems and Robotics

- This project aims to evaluate a policy through parameter synthesis in an uncertain environment where the system is modeled as an uncertain Markov Decision Process.

Optimum Motion Planning using PRM^* and RRT^*

Spring 2023

Project for COM S 576: Motion Planning for Robotics and Autonomous Systems

- This project aims to find the optimal feasible path using existing algorithms and compares existing algorithms.

Software Vulnerability Detection

Fall 2022

Project for COM S 513 - Program Analysis

- It focused on enhancing software vulnerability detection by using program slicing and transformer-based models.
- It slices a program into multiple parts so that it does not discard a program part with a vulnerability.

Predicting Loan Acceptance

Spring 2022

Project for COM S 573 - Machine Learning

- Using machine learning algorithms to predict loan acceptance using the UCI credit default dataset.

SKILLS

Programming Skills	Python, Java, C, C++, Rust, C#, Shell scripting (Bash), SQL, and Verilog
Networking Tools	DPDK
Machine Learning	Numpy, SciKit learn, Pandas, Seaborn, SciPy, and Pytorch
Formal Verification Tools	Prism, Storm, NuSMV, Spin, SMT, SAT, Z3, CTL, and LTL
DevOps & Tools	Docker, Git, Makefile, LLVM, AFL, and Klee
Robot Operating System	ROS
Vehicle Simulator	CARLA
Languages	English and Bengali (native)

REFERENCES

Song Han

Associate Professor, School of Computing
University of Connecticut, Storrs, CT
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Tichakorn (Nok) Wongpiromsarn

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Department of Computer Science
Iowa State University, Ames, IA
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